



Cairo University
 Faculty of Engineering
 Credit Hours System
Communications and Computer Engineering Department

Fall 2026

Course Code EECS 304
 Course Title Control I
 Course Instructors Dr. Ragia Badr - Dr. Nasr Antar
 Course TA Eng. Hassan El-Menier

Sheet 10: Pole Placement

Question (1):

A system is characterized by the state equation

$$\dot{\underline{x}} = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix} \underline{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(t)$$

where $u(t)$ = unit step input. Compute the solution of the state vector if the initial conditions are $x_1(0) = 1$ and $x_2(0) = -1$

Question (2):

Compute the solution of the state vector for the system

$$\dot{\underline{x}} = \begin{bmatrix} 0 & 2 \\ -3 & -5 \end{bmatrix} \underline{x} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

given that $x^T(0) = [3 \ 1]$ and $u(t)$ is a unit impulse.

Question (3):

Consider the system

$$\dot{\underline{x}} = A\underline{x} + B, \quad y = C\underline{x}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -3 & -2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [1 \ 0 \ 0]$$

- Check the controllability of the system.
- Construct a control $u(t) = -k_1x_1(t) - k_2x_2(t) - k_3x_3(t)$, such that the characteristic equation of the system has roots at -2, -5 and -6. Determine the state feedback gain vector K and the control signal $u(t)$

Question (4):

Consider the system

$$\dot{\underline{x}} = A\underline{x} + B, \quad y = C\underline{x}$$

where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [1 \quad 0 \quad 0]$$

- Check the controllability of the system.
- By using state feedback control $u(t) = -\underline{K}\underline{x}(t)$, where $\underline{K}$ is a state feedback vector, it is desired to have the closed loop poles at:

$$\lambda_1 = -2 + j3.464, \quad \lambda_2 = -2 - j3.464, \quad \lambda_3 = -5$$

Determine the state feedback gain vector $\underline{K}$ and the control signal $u(t)$.

Question (5):

Given:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 25 & 3 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = [1 \quad 1 \quad 2]$$

- Can you use output feedback with feedback gain $K = 1$ to place a pole at $s = -4$? Verify your answer and find the poles of the overall system.
- If the A matrix was modified to become $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -23 & -25 & -7 \end{bmatrix}$, can you use output feedback with feedback gain $K = 1$ to place a pole at $s = -4$? Verify your answer and find the poles of the overall system.

Hint: Output feedback means $u(t) = -Ky(t)$.